WEB-BASED APPLICATION

DEVELOPMENT
(CSE 3023)

Lab 1

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Date: 8/4/2026

Group: K1

Program: Computer Science (Software Engineering)

Lab 2 Exercise

Source Code:

ProductDAO.java

```
package com.lab.dao;

import java.sql.*;
import com.lab.model.*;
import java.util.ArrayList;
import java.util.List;

public class ProductDAO {

    //MySQL information
    private String jdbcURL = "jdbc:mysql://localhost:3307/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "";

    //Get connection
    protected Connection getConnection() {
        Connection conn = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            conn = DriverManager.getConnection(jdbcURL,
jdbcUsername, jdbcPassword);
            return conn;

        } catch (Exception e) {
            System.out.println(e);
        }
        return null;
    }

    //Add product to MySQL
    public void addProduct(Product product) {
        String sql = "INSERT INTO product (name, category, price,
quantity) VALUES (?, ?, ?, ?)";
        try (Connection conn = getConnection(); PreparedStatement
pstmt = conn.prepareStatement(sql)) {
            //Prepare sql
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());

            //Add product
            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
```

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        }
    }

    //Get product based on ID
    public Product getProductById(int id) {
        Product product = null;
        String sql = "SELECT * FROM product WHERE id =  ?";

        try (Connection conn = getConnection(); PreparedStatement
pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);

            //Get query result
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                //Extract query result
                String name = rs.getString("name");
                String category = rs.getString("category");
                double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");

                //Return product
                return new Product(id, name, category, price,
quantity);
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return product;
    }

    //Retrieve all products from MySQL
    public List<Product> getAllProducts() {
        List<Product> products = new ArrayList<>();
        String sql = "SELECT * FROM product";

        try (Connection conn = getConnection(); PreparedStatement
pstmt = conn.prepareStatement(sql)) {
            ResultSet rs = pstmt.executeQuery();

            //Iterate through every record
            while (rs.next()) {
                int id = rs.getInt("id");
                String name = rs.getString("name");
                String category = rs.getString("category");
                double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");

                //Add every product into a list
                products.add(new Product(id, name, category, price,
```

```
quantity));
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return products;
    }

    //Update exisiting record into MySQL
    public boolean updateProduct(Product product) {
        boolean rowUpdated = false;
        String sql = "UPDATE product SET name = ?, category = ?,
price = ?, quantity = ? WHERE id = ?";
        try (Connection conn = getConnection(); PreparedStatement
pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.setInt(5, product.getId());

            //Update existing record and return true if succesfully
update MySQL
            rowUpdated = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {

            e.printStackTrace();
        }
        return rowUpdated;
    }

    //Remove record from MySQL
    public boolean deleteProduct(int id) {
        boolean rowDeleted = false;
        String sql = "DELETE FROM product WHERE id = ?";

        try (Connection conn = getConnection(); PreparedStatement
pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);

            //Delete record and return true if success
            rowDeleted = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowDeleted;
    }

}
```

product.java

```
package com.lab.model;

public class Product {
    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;

    //Constructors
    public Product(int id, String name, String category, double
price, int quantity){
        this.id = id;
        this.name  = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    public Product(String name, String category, double price, int
quantity){
        this.name  = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    //Setters
    public void setId(int id){
        this.id = id;
    }

    public void setName(String name){
        this.name = name;
    }

    public void setCategory(String category){
        this.category = category;
    }

    public void setPrice (double price){
        this.price = price;
    }

    public void setQuantity(int quantity){
        this.quantity = quantity;
    }
```

```
    //Getters
    public int getId(){
        return id;
    }

    public String getName(){
        return name;
    }

    public String getCategory(){
        return category;
    }

    public double getPrice(){
        return price;
    }

    public int getQuantity(){
        return quantity;
    }
```

ViewServlet.java

```
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.util.List;

public class ViewServlet extends HttpServlet {

    private ProductDAO productDAO;

    //Initialize ProductDAO
    public void init() {
        productDAO = new ProductDAO();
    }

    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
    }

    @Override
    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        //Get all products
        List<Product> products = productDAO.getAllProducts();

        //Output
        out.println("<h2>Product List</h2>");
        out.println("<table
border='1'><tr><th>ID</th><th>Name</th><th>Category</th><th>Price</t
h><th>Quantity</th> <th>Actions</th></tr>");
        for (Product product : products) {
            out.println("<tr>");
            out.println("<td>" + product.getId() + "</td>");
            out.println("<td>" + product.getName() + "</td>");
            out.println("<td>" + product.getCategory() + "</td>");
            out.println("<td>" + product.getPrice() + "</td>");
```

```
            out.println("<td>" + product.getQuantity() + "</td>");
            out.println("<td><a href='UpdateServlet?id=" +
product.getId()
                    + "'>Edit</a> | ");
            out.println("<a href='DeleteServlet?id=" +
product.getId() + "'>Delete</a > </td>");
            out.println("</tr>");
        }
        out.println("</table>");
        out.println("<br><a href='add_product.html'>Add
Product</a>");

    }

    @Override
    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

UpdateServlet.java

```
package com.lab.controller;

import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;


public class UpdateServlet extends HttpServlet {

    private ProductDAO productDAO;

    //Initialize ProductDAO
    public void init() {
        productDAO = new ProductDAO();
    }

    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");

    }

    @Override
    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        //Get id from URL/Parameter
        String id = request.getParameter("id");

        //Get product based on current id
        Product existingProduct =
productDAO.getProductById(Integer.parseInt(id));

        //Reject if the product does not exist
        if (existingProduct != null) {
            out.println("<h2>Update Product </h2>");
            out.println("<form action='UpdateServlet'
```

```
method='POST'>");
            out.println("<input type='hidden' name='id' value='"
                    + existingProduct.getId() + "'>");

            out.println("Product Name: <input type='text'
name='name'value ='" + existingProduct.getName()+ "'
required><br><br>");
            out.println("Category: <input type='text'
name='category'value ='" + existingProduct.getCategory()+ "'
required><br><br>");
            out.println("Price: <input type='number'
name='price'value ='" + existingProduct.getPrice()+ "'
required><br><br>");
            out.println("Quantity: <input type='number'
name='quantity'value ='" + existingProduct.getQuantity()+ "'
required><br><br>");

            out.println("<input type='submit' value='Update
Product'>");
            out.println("</form>");
        } else {
            out.println("<h1>Error</h1>");
            out.print("<p>This");
            out.print(id);
            out.print(" does not exist in the system </p>");
        }

    }

    @Override
    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        //Get product info
        int id = Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price =
Double.parseDouble(request.getParameter("price"));
        int quantity =
Integer.parseInt(request.getParameter("quantity"));

        //Update product to MySQL
        productDAO.updateProduct(new Product(id, name, category,
price, quantity));

        response.sendRedirect("ViewServlet");
    }
```

```
    @Override
    public String getServletInfo() {
        return "Short description";
    }

}
```

InsertServlet.java

```
package com.lab.controller;

import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import com.lab.model.Product;
import com.lab.dao.ProductDAO;

public class InsertServlet extends HttpServlet {

    private ProductDAO productDAO;

    //Initialize ProductDAO
    public void init() {
        productDAO = new ProductDAO();
    }

    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

    }

    @Override
    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        //Get product info
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price =
Double.parseDouble(request.getParameter("price"));
        int quantity =
Integer.parseInt(request.getParameter("quantity"));

        //Add product to MySQL
        productDAO.addProduct(new Product(name, category, price,
```

```
quantity));

        response.sendRedirect("ViewServlet");
    }

    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

DeleteServlet.java

```
package com.lab.controller;

import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import com.lab.dao.ProductDAO;


public class DeleteServlet extends HttpServlet {
    private ProductDAO productDAO;

     //Initialize ProductDAO
    public void init() {
        productDAO = new ProductDAO();
    }

    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

    }

    @Override
    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        //Get product id from URL
        String id = request.getParameter("id");

        //Remove product from MySQL
        boolean success =
productDAO.deleteProduct(Integer.parseInt(id));

        //Redirect to correct page if succefully delete the product
        if (success){
            response.sendRedirect("ViewServlet");
        }else{
            //Show error
            response.setContentType("text/html");
            PrintWriter out = response.getWriter();
            out.println("<h1>Error</h1>");
            out.print("<p>Unable to delete ");
            out.println(id);
            out.print(" from the system </p>");
```

```
            out.println("<a href='ViewServlet'>Main Menu</a>");
        }
    }

    @Override
    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }


    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

add_product.html

```
<!DOCTYPE html>
<html>
    <head>
        <title>Product Management System</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    </head>
    <body>
        <h2>Add Product</h2>
        <form action="InsertServlet" method="POST">
            <label for="name">Product Name: </label>
            <input type="text" id="name" name="name"><br><br>

            <label for="category">Category: </label>
            <input type="text" id="category"
name="category"><br><br>

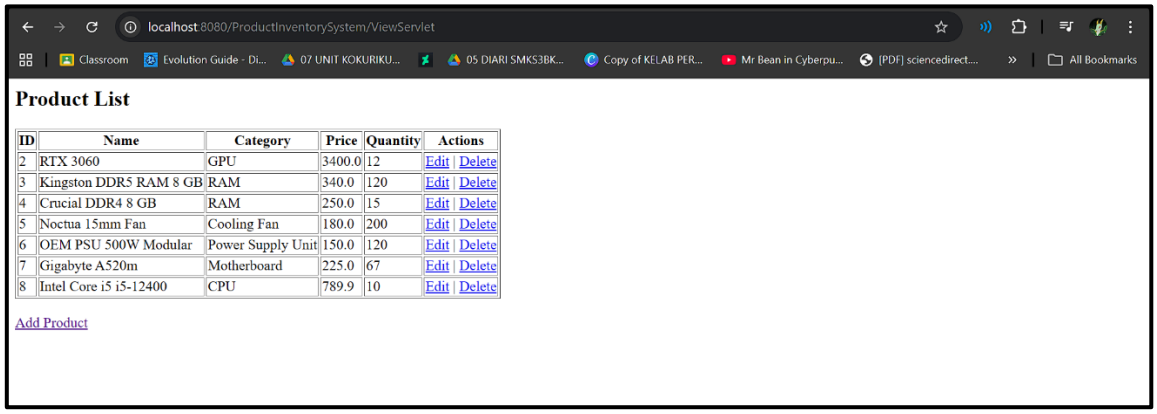
            <label for="price">Price(RM): </label>
            <input type="number" step=".01" min="0" id="price"
name="price"><br><br>

            <label for="quantity">Quantity: </label>
            <input type="number" min="0" id="quantity"
name="quantity"><br><br>

            <input type="submit" value="Add Product">
        </form>
    </body>
</html>
```

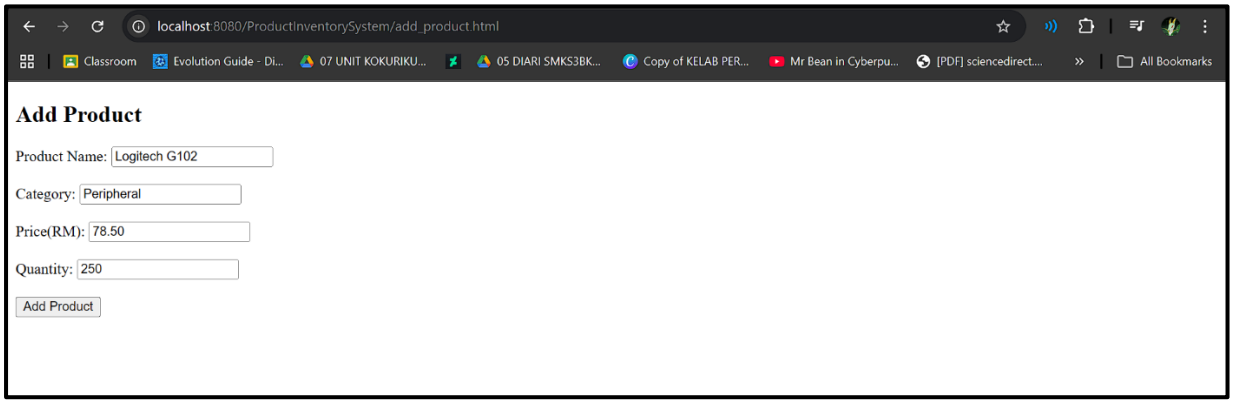
Output:

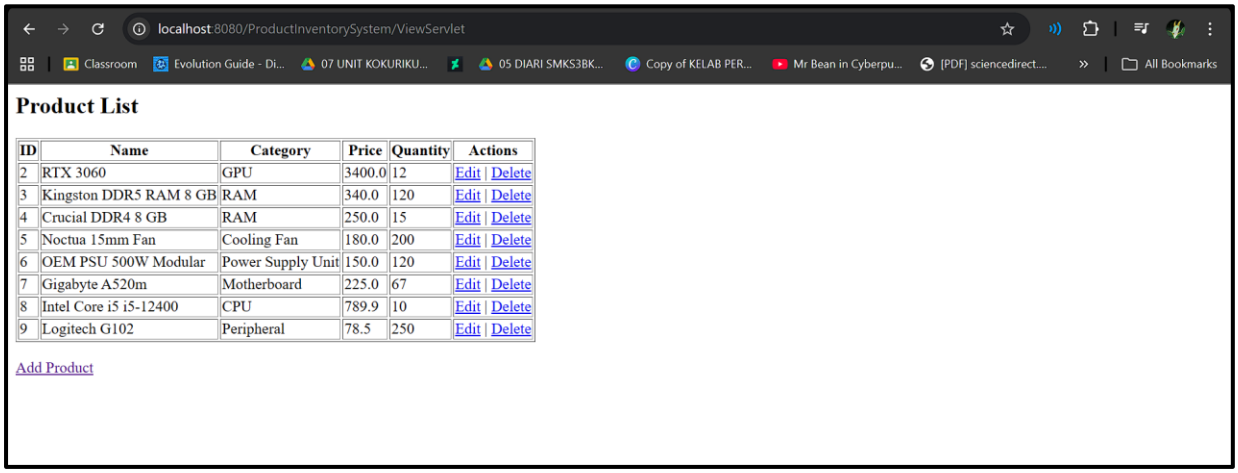
View All Products



ID	Name	Category	Price	Quantity	Actions
2	RTX 3060	GPU	3400.0	12	Edit | Delete
3	Kingston DDR5 RAM 8 GB	RAM	340.0	120	Edit | Delete
4	Crucial DDR4 8 GB	RAM	250.0	15	Edit | Delete
5	Noctua 15mm Fan	Cooling Fan	180.0	200	Edit | Delete
6	OEM PSU 500W Modular	Power Supply Unit	150.0	120	Edit | Delete
7	Gigabyte A520m	Motherboard	225.0	67	Edit | Delete
8	Intel Core i5 i5-12400	CPU	789.9	10	Edit | Delete

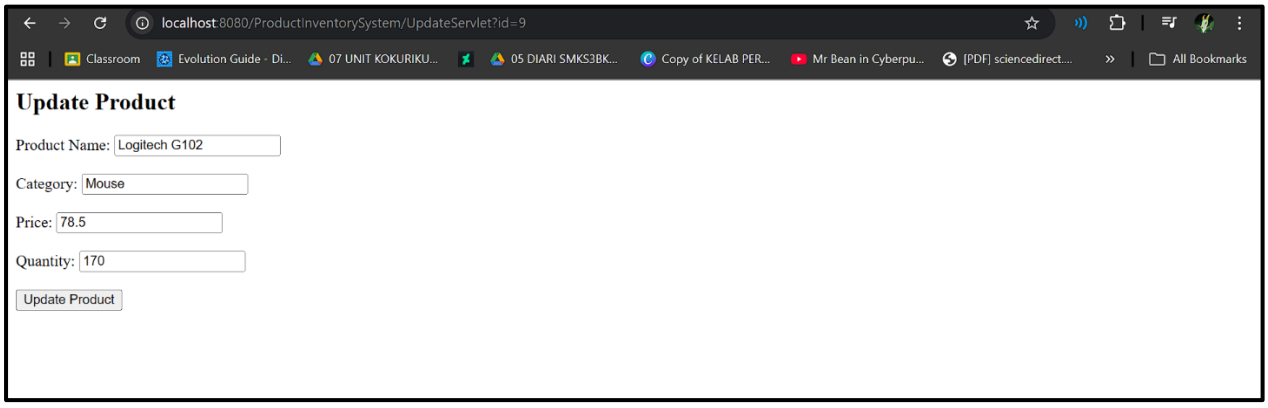
Add Product

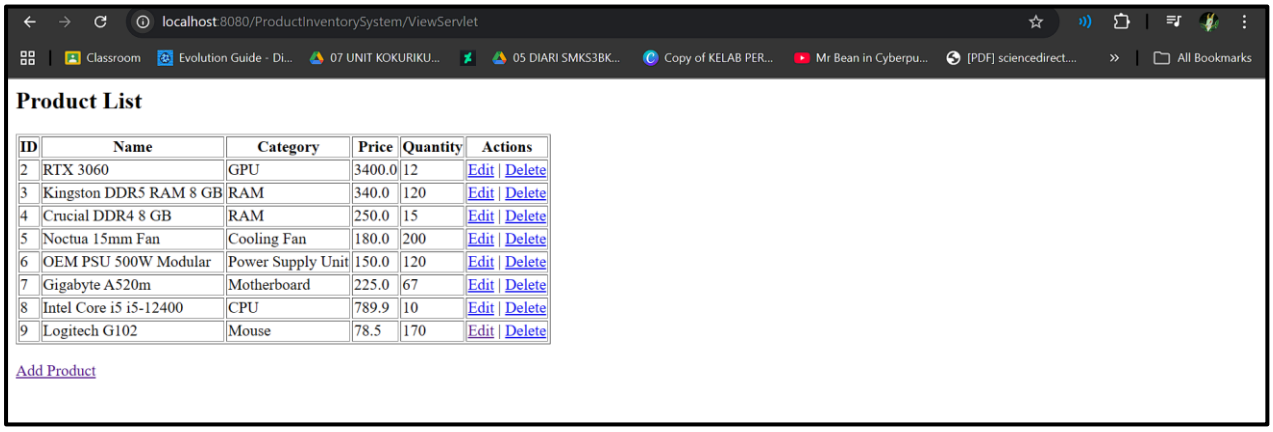




ID	Name	Category	Price	Quantity	Actions
2	RTX 3060	GPU	3400.0	12	Edit | Delete
3	Kingston DDR5 RAM 8 GB	RAM	340.0	120	Edit | Delete
4	Crucial DDR4 8 GB	RAM	250.0	15	Edit | Delete
5	Noctua 15mm Fan	Cooling Fan	180.0	200	Edit | Delete
6	OEM PSU 500W Modular	Power Supply Unit	150.0	120	Edit | Delete
7	Gigabyte A520m	Motherboard	225.0	67	Edit | Delete
8	Intel Core i5 i5-12400	CPU	789.9	10	Edit | Delete
9	Logitech G102	Peripheral	78.5	250	Edit | Delete

Update/Edit Product

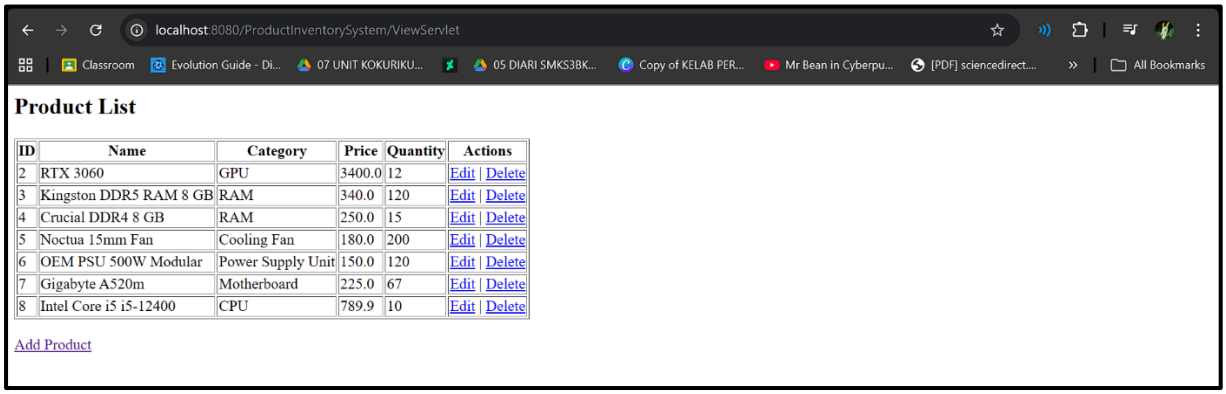




ID	Name	Category	Price	Quantity	Actions
2	RTX 3060	GPU	3400.0	12	Edit | Delete
3	Kingston DDR5 RAM 8 GB	RAM	340.0	120	Edit | Delete
4	Crucial DDR4 8 GB	RAM	250.0	15	Edit | Delete
5	Noctua 15mm Fan	Cooling Fan	180.0	200	Edit | Delete
6	OEM PSU 500W Modular	Power Supply Unit	150.0	120	Edit | Delete
7	Gigabyte A520m	Motherboard	225.0	67	Edit | Delete
8	Intel Core i5 i5-12400	CPU	789.9	10	Edit | Delete
9	Logitech G102	Mouse	78.5	170	Edit | Delete

Delete Product

ID	Name	Category	Price	Quantity	Actions
2	RTX 3060	GPU	3400.0	12	Edit | Delete
3	Kingston DDR5 RAM 8 GB	RAM	340.0	120	Edit | Delete
4	Crucial DDR4 8 GB	RAM	250.0	15	Edit | Delete
5	Noctua 15mm Fan	Cooling Fan	180.0	200	Edit | Delete
6	OEM PSU 500W Modular	Power Supply Unit	150.0	120	Edit | Delete
7	Gigabyte A520m	Motherboard	225.0	67	Edit | Delete
8	Intel Core i5 i5-12400	CPU	789.9	10	Edit | Delete
9	Logitech G102	Mouse	78.5	170	Edit | Delete



ID	Name	Category	Price	Quantity	Actions
2	RTX 3060	GPU	3400.0	12	Edit | Delete
3	Kingston DDR5 RAM 8 GB	RAM	340.0	120	Edit | Delete
4	Crucial DDR4 8 GB	RAM	250.0	15	Edit | Delete
5	Noctua 15mm Fan	Cooling Fan	180.0	200	Edit | Delete
6	OEM PSU 500W Modular	Power Supply Unit	150.0	120	Edit | Delete
7	Gigabyte A520m	Motherboard	225.0	67	Edit | Delete
8	Intel Core i5 i5-12400	CPU	789.9	10	Edit | Delete